Multi-Sensory Biophilic Design

Deeper Connection

- Participatory
  - Personal ownership
  - Communal stewardship
- Multi-sensory
- Restorative Effect
  - Stress relief
  - Mental fatigue
- Existential
  - Discomfort
  - Sublime effect
  - Awe
  - Innate fear
  - Humility
  - Friction
- Place Identity
  - Historical context
  - Genius Loci
- Non-human Ecosystem
  - Awareness & Respect
  - Agency of nature
  - Inter-subject relations

Limitations & Criticisms

- Visual emphasis
  - Tokenistic
  - Green-washing
- Superficial
  - Aesthetic only
  - Lack of ecological coherence
- Misalignment with biophilic philosophy
  - Accessibility
    - Idealised version of nature
      - green as the utopian vision
      - prioritising visual impact
      - "clean" nature
    - Controlled nature
    - Inactive (dormant) nature
    - Human-centric
- Inclusivity
  - Limited Neuro-diversity
  - Limited participatory design
    - Exclusive demographic participation
      - Need for immediate results

Sensory Design Opportunities

- Social & Community
- Nature-inspired complexity
  - Temporal connection
    - Natural cycles & rhythms
    - Loosing track of time
    - Sense of Place
  - Sensory layering
    - Gradual exposure
- Occupant Agency
- Mental & Physical Health
  - Bodily Movement
  - Attention Restoration Theory (ART)
- Discovery & Exploration
- Interaction & Response
  - Imagination
  - Surprise
- The 6 Senses

Technology

- Smart materials
  - Smart Glass
  - Blinds
  - Light
  - Smart Paint
- Robotic technology
  - Biophilic capsules
  - Presence mediation
  - Prompting & Nudging
- AI
  - Transformers
    - Biophilic Planning
  - Generative Models
    - Large Language Models
    - Visual AI
- Digital platforms
  - Participatory Tools
  - Nature awareness systems
- Mixed Reality
  - Nature enhancement
    - Mimicking Complexity
- 3D & digital fabrication
  - Hybrid crafts

Socio-economic Challenges

- Morphological constraints
  - Historical cities
  - Developing world
- Developer risk aversion
  - Liability
  - Maintenance
- Regulatory, economic, & technical constraints
  - Focus on ROI
  - Budget-driven
- Knowledge & Data limitations
  - Limited metrics
  - Lack of quantified data

Values

- Thriving eco-system
- Sustainability
- Justice & Equity
- Feminism
- Community-building
- Artistic Expression
- Inclusion & Accessability

Future Visions

- Multi-Emotional Spaces
  - Body Rythmn
  - Body Awareness
  - Emotion-specific zones
  - Mental Restoration
- Community-Centred
  - Inclusive
  - Active programming
- Technology-Enhanced Biophilia
  - Novel & speculative features
  - "Smart" & "Extra" Nature
  - Multi-sensory immersion
- Thriving Ecosystem
  - Biotope
  - Ecological corridors
  - Wildlife encounters
  - Muti-species flourishing
  - Circular Design